Prebuilt

Every printer learns from every other. No data leaves.

Every printer is wrong in its own way

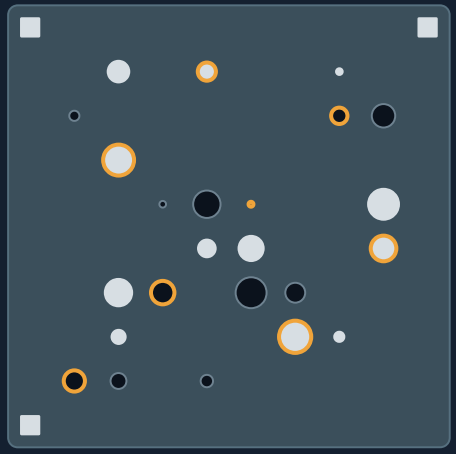
Holes print up to 250 µm small and pins up to 195 µm large (desktop SLA, CMM; Tehnički vjesnik 2026).

In medicine, that decides the fit

Crowns, surgical guides and microfluidic chips must fit first time. A crown margin allows ~120 µm.

And labs can't pool what they learn

Patient scans and process recipes can't leave, so every lab re-learns its printer from scratch.



Our test plate on a Formlabs Form Studio 1: a 1.50 mm pin printed at 1.44 mm.

HOW IT WORKS

Learn locally, share three numbers

1 Print a test plate

24 pins and holes, measured by micrometer or flatbed scanner.

2 Learn your printer

A physics model of your machine, fit locally on your own data.

3 Share three numbers

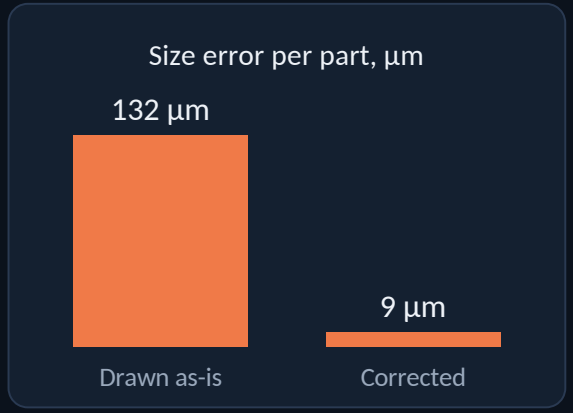
Mean and spread of 3 parameters every printer shares, logged on a receipt.

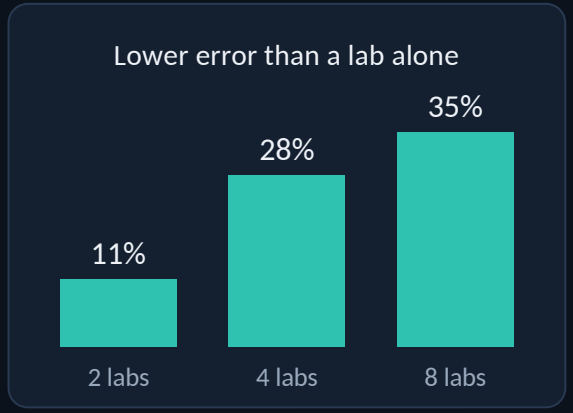
Never leaves the lab: patient scans, part geometry, measurements, machine settings. Every crossing is audited; claims were registered before the first print.

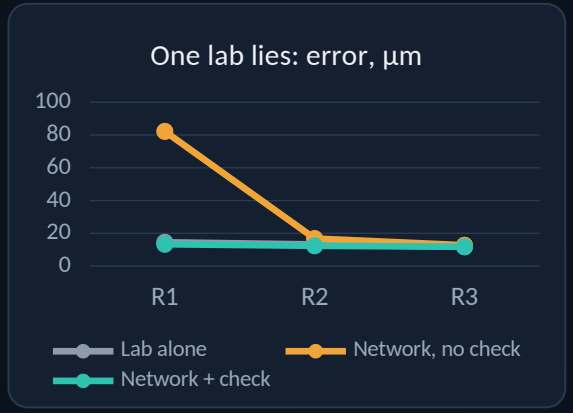
What's new: correcting printer error (Huang et al. 2015) and learning across printers (Wei et al. 2023) are proven, but pool the data. Prebuilt does it privately.

RESULTS · SIMULATION

Parts that fit, and new labs get there faster







Head to head, simulated fleets	Lower error	Fleets won	p
Network vs a lab learning alone (2 labs)	12.3%	75 / 100	<0.001
Pooling all raw data, the upper bound	12.2%	64 / 100	
Eight data-poor labs vs alone	34.7%	20 / 20	<0.001
Control: printers that share no physics	no gain	19 / 60	

WHO IT'S FOR

For anyone whose precision is a trade secret

NOW · MEDICAL AND BIO

- Hospital 3D-printing labs
- Dental labs
- Bio and microfluidics labs

NEXT · HIGH-PRECISION INDUSTRY

- Chip makers, GPU packaging
- Defense and aerospace
- MEMS and nanotech

Value grows with every member: 11% → 28% → 35% lower error at 2 / 4 / 8 labs. From microns today to nanometres next.

Print anywhere, trust the part.

github.com/matte-realize/prebuilt